Integration of vector functions

→ Let $\bar{f}(t)$ and $\bar{F}(t)$ be a vector fns of a scalar vble t s.t. $\frac{d}{dt}\bar{F}(t) = \bar{f}(t)$, then $\bar{F}(t)$ is called an integral of $\bar{f}(t)$ w.r. to t and we write $\int \bar{f}(t)\,dt = \bar{F}(t)$.

$\bar{F}(t)$ is called the indefinite integral of $\bar{f}(t)$.

→ The definite integral of $\bar{f}(t)$ b/w the limits $t=a$ and $t=b$ is written as

$$\int_a^b \bar{f}(t)\,dt = \left.\bar{F}(t)\right]_a^b = \bar{F}(b) - \bar{F}(a)$$

NOTE

1. If $\bar{f}(t) = f_1(t)i + f_2(t)j + f_3(t)k$, then

$$\int \bar{f}(t)\,dt = i\int f_1(t)\,dt + j\int f_2(t)\,dt + k\int f_3(t)\,dt.$$

∴ in order to integrate a vector fn, integrate its components

ntics

① pblm

The acceleration of a particle at time t is given by $\bar{a} = 18\cos 3t\, i - 8\sin 2t\, j + 6t\, k$. If the velocity $\bar{v}$ and displacement $\bar{r}$ be zero at $t=0$, find $\bar{v}$ and $\bar{r}$ at any pt t?

$$\bar{a} = \frac{d^2\bar{r}}{dt^2} = 18\cos 3t\, i - 8\sin 2t\, j + 6t\, k.$$

Integrating we've,

$$\vec{V} = \frac{d\bar{r}}{dt} = i\int 18\cos 3t\, dt + j\int -8\sin 2t\, dt + k\int 6t\, dt$$

$$= i \cdot 18 \cdot \frac{\sin 3t}{3} + j \cdot 8 \cdot \frac{\cos 2t}{2} + k\, 6 \cdot \frac{t^2}{2} + \vec{C}$$

$$= 6\sin 3t\, i + 4\cos 2t\, j + 3t^2 k + C$$

At $t=0$, $\bar{V}=0$

$$\therefore\ 0 + 4j + C = 0$$

$$\Rightarrow \vec{C} = -4j$$

$$\therefore \bar{V} = 6\sin 3t\, i + 4\cos 2t\, j + 3t^2 k - 4j$$

$$= 6\sin 3t\, i + 4(\cos 2t - 1)j + 3t^2 k$$

$$\vec{V} = \frac{d\bar{r}}{dt} = 6\sin 3t\, i + 4(\cos 2t - 1)j + 3t^2 k$$

Integrating again,

we've, $\bar{r} = \int 6\sin 3t\, dt\, i + j\int 4\cos(2t-1)\, dt + k\int 3t^2\, dt$

$$= -i\, 6\,\frac{\cos 3t}{3} + j\, 4\, \frac{\sin(2t-1)}{2} + k \cdot \frac{3t^3}{3}$$

$$= -2\cos 3t\, i + 2\sin(2t-1)j + t^3 k + \vec{C}$$

At $t=0, \bar{r}=0$ $\Rightarrow 0 = -2i + \vec{C}$

$$\Rightarrow \vec{C} = 2i$$

$$\therefore \vec{r} = -2\cos 3t\, i + 2\sin(2t-1)j + t^3 k + 2i$$

$$= 2(1-\cos 3t)i + 2\sin(2t-1)j + t^3 k$$

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Q. If $\vec{A}(t) = (3t^2 - 2t)i + (6t-4)j + 4t\,k$, evaluate $\int_2^3 \vec{A}(t)\,dt$?

$$\int_2^3 \vec{A}(t)\,dt = \int_2^3 \left[(3t^2-2t)i + (6t-4)j + 4t\,k\right) dt$$

$$= i\int_2^3 (3t^2-2t)\,dt + j\int_2^3 (6t-4)\,dt + k\int_2^3 4t\,dt$$

$$= i\left[t^3 - t^2\right]_2^3 + j\left[3t^2-4t\right]_2^3 + k\left(2t^2\right)_2^3$$

$$= \cancel{i[27-4] + j[27}$$

$$= i\left[27 - 9 - 8 + 4\right] + j\left[27 - 12 - 12 + 8\right] + k\left[18 - 8\right]$$

$$= 14i + 11j + 10k$$

Q. If $\vec{r}(t) = 5t^2 i + t j - t^3 k$, P.T $\int_1^2 \left(\vec{r} \times \frac{d^2 r}{dt^2}\right) dt = -14i + 75j - 15k$

$$\frac{d}{dt}\left(\vec{r} \times \frac{d\vec{r}}{dt}\right) = \frac{d\vec{r}}{dt} \times \frac{d\vec{r}}{dt} + \vec{r} \times \frac{d^2\vec{r}}{dt^2}$$

$$= \vec{r} \times \frac{d^2\vec{r}}{dt^2}$$

$$\therefore \int_1^2 \left(\vec{r} \times \frac{d^2\vec{r}}{dt^2}\right) dt = \left[\vec{r} \times \frac{d\vec{r}}{dt}\right]_1^2$$

$$\vec{r} \times \frac{d\vec{r}}{dt} = \left(5t^2 i + t j - t^3 k\right) \times \left(10t\,i + j - 3t^2 k\right)$$

$$= \begin{vmatrix} i & j & k \\ 5t^2 & t & -t^3 \\ 10t & 1 & -3t^2 \end{vmatrix}$$

$$= i\left[-3t^3+t^3\right] - j\left[-15t^4+10t^4\right] + k\left[5t^2-10t^2\right]$$

$$= -2t^3 i + 5t^4 j - 5t^2 k$$

$$\therefore \int_1^2 \left(\bar{r} \times \frac{d^2 r}{dt^2}\right) dt = \left[-2t^3 i + 5t^4 j - 5t^2 k\right]_1^2$$

$$= \left[-2t^3\right]_1^2 i + \left[5t^4\right]_1^2 j - \left[5t^2\right]_1^2 k$$

$$= -2[8-1]i + 5[16-1]j - 5[4-1]k$$

$$= -14i + 75j - 15k$$

If $\bar{r} = ti - t^2 j + (t-1)k$ & $\bar{s} = 2t^2 i + 6t\, k$.

Evaluate i) $\int_0^2 \bar{r}\cdot\bar{s}\, dt$ → 12

(ii) $\int_0^2 \bar{r}\times\bar{s}\, dt$ ⇓ $-24i - \frac{40}{3}j + \frac{64}{5}k$

Q. Given that $\vec{r}(t) = 2i - j + 2k$, when $t = 2$ and $\vec{r}(t) = 4i - 2j + 3k$ when $t = 3$. S.T

$$\int_2^3 \left(\vec{r}\cdot\frac{d\vec{r}}{dt}\right) dt = 10$$

Since $\frac{d}{dt}\bar{r}^2 = 2\vec{r}\cdot\frac{d\vec{r}}{dt}$

$$\vec{r}\cdot\frac{d\vec{r}}{dt} = \frac{1}{2}\frac{d\vec{r}^2}{dt}$$

$$\therefore \int_2^3 \vec{r}\cdot\frac{d\vec{r}}{dt} = \frac{1}{2}\int_2^3 \frac{d\vec{r}^2}{dt}\, dt = \frac{1}{2}\left[\vec{r}^2\right]_2^3$$

$$= \frac{1}{2}\left[\bar{r}^2\Big|_3 - \bar{r}^2\Big|_2\right] \quad \text{——— ①}$$

$$\bar{r}\Big|_{t=3} = 4i-2j+3k \Rightarrow \bar{r}^2\Big|_{t=3} = (4i-2j+3k).(4i-2j+3k)$$

$$= 16+4+9 = \underline{\underline{29}}$$

$$\bar{r}^2\Big|_{t=2} = 2i-j+2k \Rightarrow \bar{r}^2\Big|_{t=2} = (2i-j+2k).(2i-j+2k)$$

$$= 4+1+4 = \underline{\underline{9}}$$

$$\therefore ① \Rightarrow \int_2^3 \left(\bar{r}\cdot\frac{d\bar{r}}{dt}\right) dt = \frac{1}{2}(29-9) = \underline{\underline{10}}.$$

Line Integrals.

→ Any integral which is to be evaluated along a curve is called a line integral.

The line integral of $\vec{F}$ along C is denoted by

$$\int_C \vec{F}.d\vec{R} \quad \text{or} \quad \int_C \vec{F}.\frac{d\vec{R}}{dt}\, dt.$$

It is a scalar. It is called the tangential line integral of $\vec{F}$ along c.

→ If $\vec{F}(x,y,z) = f_1 i + f_2 j + f_3 k$ and $\vec{R} = xi + yj + zk$, then

$d\vec{R} = i\,dx + j\,dy + k\,dz$

$$\therefore \int_C \vec{F} \cdot d\vec{R} = \int_C (f_1\, dx + f_2\, dy + f_3\, dz)$$

→ If the parametric eqns of the curve C are $x = x(t),\ y = y(t),\ z = z(t)$ and $t = t_1$ at A, $t = t_2$ at B then,

$$\int_C \vec{F} \cdot d\bar{R} = \int_{t_1}^{t_2} \left(f_1 \frac{dx}{dt} + f_2 \frac{dy}{dt} + f_3 \frac{dz}{dt} \right) dt$$

→ If C is a closed curve, then the integral sign $\int_C$ is replaced by $\oint_C$.

Circulation

Let $\bar{V}$ represents the velocity of a fluid particle and C is a closed curve, then the integral $\oint_C \vec{V} \cdot d\bar{R}$ is called the circulan of $\bar{V}$ round the curve C.

If circulan $= 0 \Rightarrow \bar{V}$ is said to be irrotational in D.

Work done by a force

Q. If $\bar{F} = 3xy\,i - y^2\, j$, evaluate $\int_C \vec{F} \cdot d\vec{r}$, where C is the arc of the parabola $y = 2x^2$ from $(0,0)$ to $(1,2)$?

Let $x = t$. Then the parametric eqns of the parabola are, $x = t,\ y = 2t^2$.

At $(0,0)$ $x = 0 \Rightarrow t = 0$

$(1,2)$, $x = 1 \Rightarrow t = 1$

Let $\bar{r}$ be the position vector given by

$\bar{r} = xi + yj$

$t = \Rightarrow \bar{r} = ti + 2t^2 j$

$$\frac{d\bar{r}}{dt} = i + 4tj$$

$\vec{F} = 3xyi - y^2 j$

$= 3 \times t \times 2t^2 i - (2t^2)^2 j$

$= 6t^3 i - 4t^4 j$

$$\int_C \bar{F}.d\bar{r} = \int_C \left(\bar{F}.\frac{d\bar{r}}{dt}\right) dt = \int_0^1 (6t^3 i - 4t^4 j).(i + 4tj)\, dt$$

$$= \int_0^1 (6t^3 - 16t^5)\, dt$$

$$= \left[\frac{6t^4}{4} - \frac{16t^6}{6}\right]_0^1$$

$$= \frac{6}{4} - \frac{16}{6} = \frac{3}{2} - \frac{8}{3} = \frac{9-16}{6} = \frac{-7}{6}$$

Q A vector field is given by $\vec{F} = (\sin y)\hat{i} + x(1+\cos y)\hat{j}$. Evaluate line integral over the circular path given by $x^2 + y^2 = a^2$, $z = 0$.

The parametric equations of the circular path are $x = a\cos t$, $y = a\sin t$, $z = 0$, $0 \le t \le 2\pi$

Since $z = 0$, we can take $\vec{r} = x\hat{i} + y\hat{j}$

$\Rightarrow d\vec{r} = dx\hat{i} + dy\hat{j}$

$$\therefore \oint_C \vec{F}\cdot d\vec{r} = \oint_C [\sin y\,\hat{i} + x(1+\cos y)\hat{j}]\cdot(dx\hat{i} + dy\hat{j})$$

$$= \oint_C [\sin y\,dx + x(1+\cos y)dy].$$

$$= \oint_C \sin y\,dx + x\,dy + \cos y\cdot x\,dy.$$

$$= \oint_C [\sin y\,dx + \cos y\,x\,dy] + \oint_C x\,dy$$

$$= \oint_C d(x\sin y) + \oint_C x\,dy \qquad dy = a\cos t\,dt$$

$$= \int_0^{2\pi} d(a\cos t\,\sin(a\sin t)) + \int_0^{2\pi} a\cos t\,.\,a\cos t\,dt$$

$$= a\cos t\,\sin(a\sin t)\Big]_0^{2\pi} + \int_0^{2\pi} a^2\cos^2 t\,dt$$

$$= [a\cos t\,\sin(a\sin t)]_0^{2\pi} + a^2\int_0^{2\pi} \frac{1+\cos 2t}{2}\,dt$$

$$= [a\sin(0) - a\sin(0)] + \frac{a^2}{2}\left[t + \frac{\sin 2t}{2}\right]_0^{2\pi}$$

$$= \frac{a^2}{2}[2\pi + 0 - 0] = \underline{\underline{\pi a^2}}$$

Q. If $\vec{F} = 2y\hat{i} - z\hat{j} + x\hat{k}$, evaluate $\oint_C \vec{F}\times d\vec{r}$ along the curve $x = \cos t$, $y = \sin t$, $z = 2\cos t$ from $t = 0$ to $t = \pi/2$?

$$\vec{F} \times d\vec{r} = \begin{vmatrix} i & j & k \\ 2y & -z & x \\ dx & dy & dz \end{vmatrix}$$

$$= i(-z\,dz - x\,dy) - j(2y\,dz - x\,dx) + k(2y\,dy + z\,dx)$$

In terms of t,

$$\vec{F} \times d\vec{r} = [-2\cos t(-2\sin t)dt - \cos t(\cos t)dt]\,i + j\,[\cos t(-\sin t)dt - 2\sin t(-2\sin t)dt] + \hat{k}\,[-2\sin t(\cos t)dt + 2\cos t(-\sin t)dt)]$$

$$= (4\cos t\sin t - \cos^2 t)\,dt\,\hat{i} + j\,(-\cos t\sin t + 4\sin^2 t)\,dt\,\hat{j} + 0\,\hat{k}$$

$$\therefore \int_C \vec{F} \times d\vec{r} = \int_0^{\pi/2} (4\cos t\sin t - \cos^2 t)\,dt\,\hat{i} + (4\sin^2 t - \cos t\sin t)\,dt\,\hat{j}$$

$$= \int_0^{\pi/2} \left[2\sin 2t - \left(\frac{1+\cos 2t}{2}\right)\right] dt\,\hat{i} + \int_0^{\pi/2} \left[4\left(\frac{1-\cos 2t}{2}\right) - \frac{\sin 2t}{2}\right] dt\,\hat{j}$$

$$= \left[-\frac{2\cos 2t}{2} - \frac{1}{2}t - \frac{\sin 2t}{4}\right]_0^{\pi/2} \hat{i} + \left[2\left(t - \frac{\sin 2t}{2}\right) + \frac{\cos 2t}{4}\right]_0^{\pi/2} \hat{j}$$

$$= \left[1 - \pi/4 + 1\right] \hat{i} + \left[2(\pi/2 - 0) + -1/4 - \frac{1}{4}\right] \hat{j}$$

$$= \left(2 - \frac{\pi}{4}\right)\hat{i} + \left(\pi - \frac{1}{2}\right)\hat{j}$$

Q. Find the work done by the force $\vec{F} = x\hat{i} - 2\hat{j} + 2y\hat{k}$ in the displacement along the closed path C consisting of the segments C_1, C_2 and C_3 where

on C_1, $0 \le x \le 1$, $y = x$, $z = 0$

on C_2, $0 \le z \le 1$, $x = 1$, $y = 1$

on C_3, $1 \ge x \ge 0$, $y = z = x$

Total work done

$$= \oint_C \vec{F}\cdot d\vec{r} = \oint_C (x\hat{i} - z\hat{j} + 2y\hat{k})\cdot(dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$= \oint_C (x\,dx - z\,dy + 2y\,dz)$$

$$= \oint_{C_1} x\,dx - z\,dy + 2y\,dz + \oint_{C_2} x\,dx - z\,dy + 2y\,dz + \oint_{C_3} x\,dx - z\,dy + 2y\,dz$$

$$= W_1 + W_2 + W_3.$$

On C_1, $W_1 = \int_0^1 x\,dx - 0 + 2y \times 0 = \int_0^1 x\,dx = \frac{x^2}{2}\Big]_0^1 = \frac{1}{2}$

On C_2, $W_2 = \int_0^1 dx^0 - z \times 0 + 2\,dz = 2\int_0^1 dz = 2z\Big]_0^1 = 2$

On C_3, $W_3 = \int_1^0 x\,dx - x\,dx + 2x\,dx = 2\int_1^0 x\,dx = x^2\Big]_1^0 = -1$

$\therefore$ Total work done

$= W_1 + W_2 + W_3 = \frac{1}{2} + 2 - 1 = \frac{3}{2}$